

ES 666: Computer Vision

Assignment 2 — Find Waldo (Helper Guide)

The one book to have open

Richard Szeliski, *Computer Vision: Algorithms and Applications*, 2nd edition. The author hosts the full PDF for free:

<https://szeliski.org/Book/>

- Chapter 3, *Image Processing* — convolution, separable filters, padding, the median filter. Everything in Task 1.
- Chapter 7, *Feature Detection and Matching* — Harris, scale space, descriptors, the ratio test. Tasks 3 and 4.

Task 1 — filtering and quality

- Z. Wang et al., *Image Quality Assessment: From Error Visibility to Structural Similarity* (2004) — the paper SSIM comes from, and the source of the C_1 , C_2 constants you are told to use:
<https://www.cns.nyu.edu/pub/eero/wang03-reprint.pdf>
- Wikipedia, *Structural similarity index measure* — a shorter route to the same formula:
https://en.wikipedia.org/wiki/Structural_similarity_index_measure
- Wikipedia, *Peak signal-to-noise ratio*:
https://en.wikipedia.org/wiki/Peak_signal-to-noise_ratio
- OpenCV, *Smoothing Images* — useful for seeing what each filter does to a picture. Read it for the pictures; do not call the functions:
https://docs.opencv.org/4.x/d4/d13/tutorial_py_filtering.html
- NumPy, *Broadcasting* — if the `einsum` in `convolve2d` is fighting you, the problem is usually here:
<https://numpy.org/doc/stable/user/basics.broadcasting.html>

Task 2 — correlation and integral images

- J. P. Lewis, *Fast Normalized Cross-Correlation* — short, direct, and it is exactly the derivation Task 2.3 asks you to implement:
<http://scribblethink.org/Work/nvisionInterface/nip.html>
- Wikipedia, *Summed-area table* — the four-lookup trick, with the picture that makes it obvious:
https://en.wikipedia.org/wiki/Summed-area_table
- P. Viola and M. Jones, *Rapid Object Detection using a Boosted Cascade of Simple Features* (2001) — where integral images became standard practice in vision:
<https://www.cs.cmu.edu/~efros/courses/LBMV07/Papers/viola-cvpr-01.pdf>

- NumPy, `sliding_window_view` — read the note about it returning a *view*, which is the whole reason Task 2.3 runs at all:
https://numpy.org/doc/stable/reference/generated/numpy.lib.stride_tricks.sliding_window_view.html
- NumPy, `einsum`:
<https://numpy.org/doc/stable/reference/generated/numpy.einsum.html>

Task 3 — corners, scale space and descriptors

- C. Harris and M. Stephens, *A Combined Corner and Edge Detector* (1988) — four pages, and the $\det -k$ trace² response is on the third:
<https://www.bmva.org/bmvc/1988/avc-88-023.pdf>
- D. Lowe, *Distinctive Image Features from Scale-Invariant Keypoints* (IJCV 2004) — the source for the DoG ladder, the orientation histogram, the $4 \times 4 \times 8$ descriptor and the ratio test. Tasks 3.3 through 4.2 are a simplified version of this paper:
<https://www.cs.ubc.ca/~lowe/papers/ijcv04.pdf>
- Wikipedia, *Harris corner detector* and *Scale-invariant feature transform* — for orientation before reading the papers:
https://en.wikipedia.org/wiki/Harris_corner_detector
https://en.wikipedia.org/wiki/Scale-invariant_feature_transform
- T. Lindeberg, *Scale-space theory: A basic tool for analysing structures at different scales* — background for Task 3.7, on why the ladder is a trade and not just a parameter:
<https://people.kth.se/~tony/abstracts/Lin94-SI-abstract.html>

Task 4 — matching under mostly-wrong data

- Lowe (2004) again — Section 7 is the ratio test, and Section 7.1 is the pose-clustering idea that Tasks 4.3 and 4.4 are built on.
- D. Ballard, *Generalizing the Hough Transform to Detect Arbitrary Shapes* (1981) — “let every local observation vote for a global answer” is this paper, and it is what `predict_centres` plus `consensus_centre` amounts to:
<https://www.sciencedirect.com/science/article/abs/pii/0031320381900091>
- M. Fischler and R. Bolles, *Random Sample Consensus* (1981) — the standard answer to “most of my data is wrong”. You are not asked to implement it, but Task 4.6 is really asking why something like it is necessary:
<https://www.sri.com/wp-content/uploads/2021/12/ransac-publication.pdf>
- Wikipedia, *Random sample consensus*:
https://en.wikipedia.org/wiki/Random_sample_consensus

If you want to go further

- First Principles of Computer Vision (Shree Nayar, Columbia) — short, very clear video lectures covering most of this assignment:
<https://fpcv.cs.columbia.edu/>
- James Hays, *Computer Vision* (Georgia Tech) — slides and assignments in the same tradition as this one:
<https://faculty.cc.gatech.edu/~hays/compvision/>